

SentinelMail: Benchmarking Multi-Model NLP Ensembles for Sensitive-Sentence Classification

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1 Introduction

Sensitive-data leakage through organizational email is a long-standing security challenge, and the growing use of generative AI adds a new channel in which users may expose personal, financial, or confidential content in free-form text, extending the traditional Data Loss Prevention (DLP) problem. Detection is difficult because sensitive information has no uniform textual form: structured identifiers such as credit-card or social-security numbers follow recognizable patterns, while other disclosures depend on linguistic context or appear in modified formats (the same phone number written as a continuous digit sequence or separated by spaces and hyphens), and sensitive concepts may be mentioned in instructions, negations, or hypothetical scenarios without revealing actual information. No single detection approach handles all of these forms reliably.

In this work, we formulate sensitive-information detection as a multi-label classification problem over short texts, distinguishing benign content, general PII, financial information, and confidential information. We evaluate four approaches with different capabilities: a rule-based detector combining regular expressions and named-entity recognition for known structured patterns, a BiLSTM with character-level convolutional features for variations in the internal structure of sensitive tokens, and two fine-tuned transformer classifiers (BERT and RoBERTa) that model the surrounding linguistic context.

While email DLP is the motivating application, it is not the evaluated task: all quantitative results are obtained on the `ai4privacy/pii-masking-300k` dataset [2], a corpus of short synthetic PII-bearing sentences that lacks the structure of real email (subject lines, threading, quoting, signatures). Our results therefore measure sensitive-*sentence* classification, an external-validity limitation we return to in the Discussion.

Our study is organized around two research questions:

- **RQ1:** Does a multi-model ensemble outperform the best single model for sensitive-content classification?
- **RQ2:** Does model complementarity vary across sensitivity categories (PII vs. financial vs. confidential)?

To answer these questions, we combine the four models into an ensemble that averages their per-category probabilities under a separate set of learned weights for each category. This weighting scheme is standard in the classifier-combination literature [9, 10], and we do not present it as new. The contribution lies elsewhere: (i) a rigorously significance-gated empirical finding — the ensemble does beat the best single model, with a delta whose 95% confidence interval excludes zero under a paired bootstrap and survives a Holm–Bonferroni correction against every member, but the gain is practically small (a few tenths of a macro-F1 point), and we report effect size next to significance so a tiny-but-real gain is never dressed up as a large one; (ii) a mechanistic explanation of *why* the gain is small — a leave-one-model-out ablation shows the fitted ensemble’s reliance is concentrated on a single member rather than cleanly split across categories, inherently limiting the headroom over that member alone; and (iii) a reusable, honesty-first evaluation methodology for judging ensemble claims — significance gate, multi-comparison panel, ablation, operating-point view, and cost accounting.

1.1 Related Work

Sensitive-information detection has traditionally relied on regular expressions, dictionaries, and named-entity recognition: rule-based de-identification achieved high recall on protected health information in clinical text [14], pattern-matching methods dominated early de-identification research [12], and modern tools such as Microsoft Presidio [13] continue this line by combining regular-expression recognizers with NER. Because predefined patterns miss context-dependent or irregular disclosures, later work framed enterprise DLP as supervised text classification [6], applied deep learning to context-dependent sensitive information [1], combined regular expressions with a BERT-BiLSTM-Attention extractor [5], and used transformers [3, 11] for sensitive-data detection, including BERT-based anonymization of clinical text [4].

Combining classifiers to exploit complementary behavior is a classical topic: weighted soft voting and linear opinion pooling date back to the multiple-classifier-systems literature [9], per-class combiner weights are well established [10], more expressive alternatives include stacked generalization [16] and mixture-of-experts gating [8], and in multi-label settings weighted ensembles can assign per-label importance to individual classifiers [17]. Our fusion rule is an instance of per-class weighted soft voting; what we add is an empirical comparison of pattern-aware, character-aware, and context-aware models within a common sensitive-text classification framework, using the learned per-category weights and ablations to characterize complementarity across sensitivity categories.

2 Methodology

2.1 General Approach

Sensitive information can appear as a recognizable structured pattern, an irregular or obfuscated sequence, or a disclosure whose sensitivity depends on the surrounding context. We therefore evaluate four models — a rule-based detector, a BiLSTM with character-level features, and two fine-tuned transformers (BERT and RoBERTa) — each predicting, for an input text, whether it is benign and whether it contains general personally identifiable information (PII), financial information, or confidential information. Since the models capture different textual signals, their predictions are combined using category-specific confidence-weighted late fusion, and the individual models and the ensemble are compared using per-category metrics, confidence intervals, statistical significance testing, error analysis, and model ablation.

2.2 Data Preparation and Label Construction

The dataset provides each text together with a `privacy_mask` containing the sensitive entities found in it. We convert these entity-level annotations into a four-dimensional text-level label vector $\mathbf{y} = [y_{benign}, y_{PII}, y_{financial}, y_{confidential}]$: a text is benign when it contains no annotated sensitive entity; any sensitive entity activates the PII label; financial identifiers (social-security numbers and card-related entities) additionally activate the financial label; and passwords and identity-document identifiers (passport, ID card, driver’s licence) activate the confidential label. The labels are hierarchical — financial and confidential are subsets of PII, and benign is mutually exclusive with all sensitive categories — so a single text may belong to more than one sensitive category.

We note that the labels are *derived* from the dataset’s entity annotations rather than independently annotated: the ground truth is a deterministic function of the entity mask, and part of what the models learn is to recover this mapping. This is appropriate for comparing model families on a shared task, but it should be kept in mind when interpreting absolute scores, and it partly explains why pattern-based models are competitive on categories defined by regex-shaped entities.

After filtering to English, the data contain 29,908 training and 7,946 validation examples. We reserve the dataset’s `validation` split exclusively for final reporting (TEST) and carve a seeded 10% slice out of the training split (`train_holdout`) as the development set (DEV) for model selection, ensemble-weight learning, and threshold tuning. All models share the same preprocessing and label-construction functions, so their predictions are evaluated against identical ground truth.

2.3 Individual Models

2.3.1 Rule-Based Detector

The rule-based detector is the pattern-aware component. It uses Microsoft Presidio [13], which combines regular-expression recognizers with spaCy named-entity recognition, and identifies entities such as email addresses, phone numbers, credit-card numbers, bank identifiers, social-security numbers, passports, and driver’s licences. Because stock Presidio has no recognizers for several confidential entity types in our schema, we added three custom recognizers covering password-like secrets, generic identity-document numbers, and classification-marker vocabulary; detected entities are mapped to the four-label schema through the same entity mapping used for ground-truth construction. The model requires no training and provides transparent detections, but is expected to be less effective when sensitive information is obfuscated, implicit, or context-dependent.

2.3.2 BiLSTM with Character-Level Features

The BiLSTM model is the character-aware component, intended to recognize variations that fixed rules miss — unusual separators, formatting changes, and out-of-vocabulary tokens. Each token combines a frozen pre-trained 100-dimensional GloVe word embedding [15] with a character-level representation produced by convolutional filters of different sizes, which learn local structural patterns within words and identifiers. The concatenated representation passes through a two-layer bidirectional LSTM [7], mean pooling over non-padding positions, and a linear layer with four sigmoid outputs giving independent per-label probabilities. The model is trained for 20 epochs with batch size 64 and learning rate 10^{-3} , keeping the checkpoint with the best validation loss.

2.3.3 Fine-Tuned Transformers: BERT and RoBERTa

BERT [3] and RoBERTa [11] are the context-aware components, intended to capture information that fixed patterns cannot express, such as implicit disclosures and mentions in instructions, negations, or hypotheticals. Texts are tokenized with each model’s native tokenizer (**bert-base-uncased** WordPiece and **roberta-base** BPE) and truncated to 512 tokens; the contextual representation of the initial classification token feeds a linear layer producing four sigmoid-activated label probabilities. To reduce computational cost, the embedding layer and first ten transformer blocks are frozen and only the final two blocks and the classification head are fine-tuned — trading a small amount of accuracy for a training footprint feasible on our hardware. Training uses binary cross-entropy with logits, positive-class weighting, label smoothing, gradient clipping, and a learning-rate schedule with warmup.

2.4 Confidence-Weighted Late Fusion

Each model produces a probability $p_{m,l}(x)$ for model m , label l , and input text x . The ensemble combines these probabilities separately for each label:

$$p_l^{ensemble}(x) = \frac{\sum_m w_{m,l} p_{m,l}(x)}{\sum_m w_{m,l}}, \quad (1)$$

where $w_{m,l}$ is the weight assigned to model m for label l . Unlike uniform averaging, this lets the same model have different importance across sensitivity categories — e.g., the rule-based detector for structured financial identifiers, a transformer for context-dependent PII.

The category-specific weights are fit on the DEV split by a per-label grid search refined with Nelder–Mead optimization of the true per-label F1. Per-label decision thresholds are tuned on the same DEV split and applied unchanged when reporting on TEST. We deliberately chose this simple linear pooling over stacking [16] or input-conditioned gating [8]: with only four members it is less prone to overfitting the small development split, and — central to RQ2 — the learned weights remain directly interpretable as a measure of how much each category relies on each model family.

2.5 Evaluation Protocol

All models are evaluated with a shared framework on the untouched TEST split. We report per-category precision, recall, and F1, macro-F1, and ROC-AUC, with a 95% bootstrap confidence interval for macro-F1 (1,000 resamples, fixed seed). For RQ1, a paired bootstrap over TEST examples tests whether the ensemble’s macro-F1 gain over the best single model is significant, extended to a pairwise panel against *every* member with a Holm–Bonferroni correction so the verdict is not an artifact of picking one comparison; effect sizes (deltas with confidence intervals) are reported alongside p-values throughout, and model stability is assessed with 5-fold cross-validation. We also analyze errors associated with negation, hypothetical language, implicit disclosure, and obfuscation — descriptively only, since the dataset provides no controlled contrast set, so no robustness claim is attached. Finally, a leave-one-model-out ablation, together with the learned per-category weights, is used to answer RQ2: whether each model provides a distinct contribution, and to which categories.

3 Experimental results

3.1 Settings

All numbers below are computed on the untouched TEST split (7,946 English examples), with ensemble weights and decision thresholds fit on DEV and applied unchanged: 0.80 (benign), 0.10 (PII), 0.50 (financial), 0.55 (confidential). Throughout, *headline macro-F1* averages F1 over the three informative categories (PII, financial, confidential); **benign** is reported per label but excluded

Model	benign	PII	financial	confid.	macro-F1
Majority prior	0.000	0.939	0.000	0.000	0.313
Rule-based	0.605	0.943	0.656	0.807	0.802
BiLSTM	0.856	0.982	0.902	0.945	0.943
BERT	0.929	0.991	0.922	0.970	0.961
RoBERTa	0.928	0.990	0.948	0.977	0.972
Ensemble	0.927	0.991	0.954	0.979	0.974

Table 1: Per-label F1 and headline macro-F1 (PII/financial/confidential) on TEST. 95% bootstrap CIs for the two leading systems: RoBERTa [0.968, 0.975], ensemble [0.971, 0.978].

vs. member	member macro-F1	delta	95% CI	Holm p
RoBERTa	0.972	+0.0026	[+0.0006, +0.0048]	0.006
BERT	0.961	+0.0132	[+0.0103, +0.0162]	< 0.001
BiLSTM	0.943	+0.0316	[+0.0281, +0.0354]	< 0.001
Rule-based	0.802	+0.1721	[+0.1645, +0.1799]	< 0.001

Table 2: Paired-bootstrap panel: ensemble macro-F1 delta over each member on TEST, Holm-corrected across the four tests. Every delta CI excludes zero.

from the macro, since it is the complement of PII and would double-count the same decision. Confidence intervals use bootstrap resampling (1,000 resamples, fixed seed).

3.2 Main Comparison

Table 1 compares all systems under identical thresholds and ground truth. A majority-prior baseline (predict the most frequent label configuration) is included for orientation.

The expected capability ordering holds: the rule-based detector is strong on structured PII but weak where patterns are irregular or contextual, the character-aware BiLSTM closes much of that gap, and the transformers are strongest, with the ensemble attaining the best headline macro-F1. Ensemble per-label ROC-AUC is ≥ 0.99 for every category (macro AUC 0.994).

RQ1 verdict. A paired bootstrap over TEST examples compares the ensemble’s headline macro-F1 (0.9744) against the best single model, RoBERTa (0.9717): the delta is +0.0026 with 95% CI [+0.0006, +0.0048] and $p = 0.006$. **The ensemble outperforms the best single model, but by a practically small margin.** Table 2 extends this to a Holm–Bonferroni-corrected panel against every member: each delta remains significant, with magnitude scaling as the member gets weaker. At $N \approx 8k$ the paired bootstrap resolves tiny-but-real differences, so “significant” here means *reliably positive*, not *large*.

As a stability check, 5-fold cross-validation of the neural members yields headline macro-F1 of 0.964 ± 0.002 (RoBERTa), 0.959 ± 0.003 (BERT), and

Model	benign	PII	financial	confid.
BERT	0.10	0.10	0.10	0.00
RoBERTa	0.70	0.70	0.30	0.60
BiLSTM	0.10	0.20	0.30	0.30
Rule-based	0.10	0.00	0.30	0.10

Table 3: Learned per-category fusion weights (DEV-fit). Each column is normalized at fusion time. RoBERTa carries the dominant weight on three of four categories; financial is the only category with weight spread across several members.

Configuration	PII	financial	confid.	macro-F1
Full ensemble	0.991	0.954	0.979	0.974
– BERT	0.990	0.951	0.979	0.973
– RoBERTa	0.990	0.943	0.944	0.959
– BiLSTM	0.991	0.948	0.977	0.972
– Rule-based	0.991	0.939	0.979	0.970
RoBERTa alone	0.990	0.948	0.977	0.972

Table 4: Leave-one-member-out ablation on TEST (per-label F1 and headline macro-F1).

0.941 ± 0.006 (BiLSTM), consistent with the single-split ranking in Table 1.

3.3 Per-Category Reliance (RQ2)

Table 3 shows the learned fusion weights, and Table 4 the leave-one-member-out ablation.

The learned weights (Table 3) look like per-category specialization: RoBERTa heavy on benign, PII, and confidential, with financial spread across three members. The ablation tells a different story. Removing RoBERTa costs about 1.5 macro-F1 points ($0.974 \rightarrow 0.959$; benign $0.927 \rightarrow 0.765$ per-label), while removing *any other* member barely moves the score (all remaining configurations stay at 0.970 or above). PII stays at ≈ 0.990 under every single removal, and financial — the one category with spread weights — shows only mild reliance: dropping the BiLSTM leaves financial at 0.948, essentially tied with RoBERTa alone. The answer to RQ2 is therefore a **qualified no**: the fitted ensemble’s real reliance is concentrated on RoBERTa almost everywhere, not cleanly differentiated by category (note this measures reliance under the fixed DEV-fit weights, not intrinsic complementarity). This concentration also explains the RQ1 result mechanistically: an ensemble dominated by one member is essentially that member plus a little, so the headroom to beat it is inherently small — exactly the reliably-positive-but-tiny gain the paired bootstrap measures.

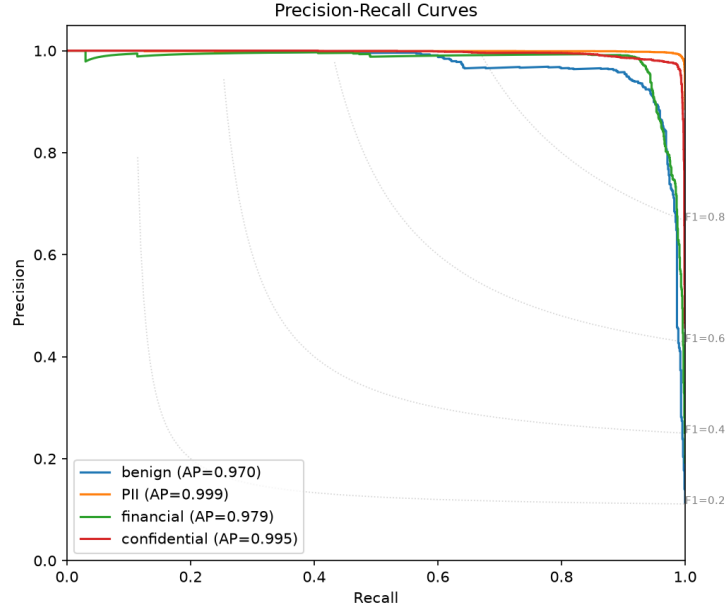


Figure 1: Ensemble precision–recall curves per category on TEST (dotted lines are iso-F1 contours). DLP deployment operates at the far right of these curves (recall ≥ 0.99), where the benign and financial curves drop steeply.

3.4 DLP Operating Point

Headline macro-F1 uses F1-optimal thresholds, but a deployed DLP system runs at high recall, since a missed sensitive item costs far more than a false alarm. Figure 1 shows the ensemble’s precision–recall curves per category. At a recall floor of 0.99, the ensemble attains the best per-category precision of any system on all three informative categories — 0.993 (PII), 0.621 (financial), 0.964 (confidential) — each at least matching the best single model; on benign, BERT (0.489) beats the ensemble (0.427). Three caveats keep this a supporting qualifier rather than an independent claim: the recall-floor thresholds are selected on the TEST split itself (optimistic upper bounds), the edge is not significance-tested, and the benign and financial values sit in low-precision regions where estimates are noisy.

4 Discussion

RQ1: yes, but small. The ensemble does outperform a single fine-tuned RoBERTa, and the win holds against every member under a Holm correction — but the gain over the transformers is a few tenths of a macro-F1 point, purchased at roughly $2\times$ the inference cost. Statistical significance here does

not imply operational significance: reported honestly, the finding is a gain that is reliably positive but practically small.

RQ2: a qualified no. The learned weights suggest per-category specialization, but the ablation shows the fitted ensemble’s real reliance is concentrated on RoBERTa almost everywhere — which is also the mechanism behind RQ1’s small margin. The one place the ensemble shows a broader edge, best per-category precision on all informative categories at the recall ≥ 0.99 operating point, is reported with its caveats (test-split thresholds, no significance test) as a supporting qualifier rather than an independent result.

5 Code

The complete implementation — data preparation, the four models, the fusion and evaluation packages, and notebooks 01–07, whose novelty analysis generates every number and figure in Section 3 — is available at:

<https://github.com/erel213/nlp-final-project>.

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